ComNetX: Local Hierarchical Adaptation for Batched Dynamic Community Detection

GRIGORIY BOKOV[1], ALEKSANDR KONOVALOV[1], ANNA UPOROVA[1], ALEXANDER DROBYSHEV[1], and IAROSLAV EGOROV[1]

[1]AI Research Center, Lomonosov Moscow State University, 1 Leninskie Gory, Moscow 119991, Russian Federation

Corresponding author: Grigoriy Bokov (e-mail: grigoriy.bokov@math.msu.ru).

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ABSTRACT Rerunning a community detector on a dynamic graph after every batch of edge updates can be expensive. ComNetX adapts existing detectors by selecting communities that intersect a neighborhood of the affected endpoints. It contracts vertex groups defined by a stored hierarchy, runs the detector on the resulting weighted graph, and maps the partition back to the original vertices. The first pass updates the reported finest partition. Subsequent passes update the coarser partitions while preserving nesting. On the selected induced subgraph, contraction preserves modularity for partitions that keep each contracted group intact. We evaluate update streams constructed from six real graphs. On five graphs initialized from approximately 99.9% of the stored records, warm-started Leiden on the full graph took 6.60 to 98.72 times as long as ComNetX with Leiden. Absolute differences in final modularity ranged from 0.0035 to 0.0213. On PubMed and ArxivMath, the first-pass update regions averaged 64% and 79% of the vertices, respectively, while the quotient graphs averaged less than 1% of the original vertex count. Over 500 updates covering the final 10% of the records on PubMed and ArxivMath, the corresponding time ratios were 24.46 and 72.67, respectively, and the absolute modularity difference stayed below 0.008 at every update. Speedups were smaller when the graphs grew from singleton partitions. With a spectral detector and random features, ComNetX was faster in all 20 paired runs under the respective cluster-count rules for the full and local graphs.

INDEX TERMS community detection, dynamic graphs, graph clustering, hierarchical graph contraction, incremental graph analytics

I. INTRODUCTION†

COMMUNITY structure helps characterize communication networks [1], track the evolution of social groups [2], and predict popularity in dynamic networks [3]. As edges change, maintaining a partition requires deciding which assignments to revisit [4]. Even a small batch can trigger a costly full-graph computation with a modularity heuristic such as Louvain or Leiden [5], [6], or with an attributed method that constructs features or trains a model [7]–[9].

A static detector can start afresh on every snapshot or use the preceding partition, as in the unseeded and seeded Grappolo schemes [10]. Incremental methods save work by screening vertices or changing how the optimizer processes them [11]–[14]. These rules depend on the underlying objective or optimizer. ComNetX instead changes the graph supplied to an existing detector. Its update region contains whole communities, and its stored hierarchy determines which vertices the detector can reassign individually. We make three contributions.

1) We construct a weighted quotient of each update region and lift the detector's result to the original vertices. The same procedure supports topology-only detectors and feature-aware detectors with aggregated attributes.
2) We prove that the update preserves nesting, apply the standard modularity-preserving contraction identity to each induced subgraph, and derive time and memory bounds, including the graph-wide worst case.
3) On streams from six real graphs, we compare quality and processing time against full Leiden recomputation and the Dynamic Frontier variant of Leiden (DF-Leiden) [14]. The evaluation includes 500-update sequences, graph growth, batch subdivision, and a feature-aware detector.

II. RELATED WORK†

A. COMMUNITY DEFINITIONS, OBJECTIVES, AND EVALUATION

Community definitions include criteria based on internal and external connectivity [15]–[18]. Modularity compares within-community edge weight with a degree-preserving null model

[19], [20]. Normalized cut measures separation relative to group volume [21], the map equation uses random-walk compression [22], and stochastic blockmodels specify edge probabilities between latent groups [23].

For modularity optimization, Louvain alternates local moves and aggregation [5]. Leiden adds refinement to address poorly connected communities [6]. Other static methods use shared-memory parallelism [24], [25], fast label propagation [26], modularity-guided label updates [27], greedy modularity disassembly [28], or iterated local moves [29].

Modularity has resolution effects and many near-optimal partitions [30], [31]. Algorithm rankings also depend on graph structure and evaluation design [32]. Because reference class labels can reflect a different organizing principle from structural communities [33], [34], we report modularity and normalized mutual information (NMI) with these labels separately [35].

B. INCREMENTAL AND PARALLEL COMMUNITY DETECTION

Incremental methods maintain partitions across snapshots [4] using batch-update rules [36], dynamic label propagation [37], [38], or modularity-based updates [11], [39]. Delta-Screening uses modularity tests to select vertices for reconsideration [12]. C-Blondel compresses previously detected communities into supernodes and applies Louvain to the compressed graph [40]. The dLocal algorithm of Görke et al. retains a hierarchy of contracted graphs and revisits groups affected by changes at the level below [41]. ComNetX also reuses a hierarchy, but selects the update regions at all levels before the first detector call. It then uses the updated finer partition to construct the next quotient.

Dynamic-frontier methods expand the set of vertices reconsidered by the optimizer [13], [42]. Parallel approaches include the shared-memory DyComPar algorithm [43] and the distributed DyG-DPCD algorithm [44].

For Leiden, Sahu et al. combine affected-vertex processing with selective refinement and changes to community renumbering and aggregation [14]. Subsequent work tracks community identifiers and uses accumulated changes in community edge weight to guide refinement [45]. LD-Leiden repairs a stored hierarchy around an affected frontier [46]. HIT-Leiden maintains connected components and hierarchical community structures to restrict the affected range [47]. These approaches modify the Leiden optimizer. ComNetX maintains its own hierarchy and can use any compatible detector to partition a weighted quotient.

C. TEMPORAL MODELS AND LOCAL OUTPUT REQUIREMENTS

NOME balances snapshot quality and temporal smoothness [48]. Kong et al. use TopoReg to encourage consistent inter-community structure across snapshots in a matrix factorization model [49]. Dynamic stochastic blockmodels use a state-space model [50], and flow stability accounts for temporal edge ordering [51]. Other approaches model community evolution through the Matthew effect [52] or track communities through modularity [53]. Continuous-time L-modularity avoids choosing snapshot boundaries [54]. Further work addresses sparse temporal observations [55] and random-walk snapshot clustering [56]. ComNetX updates a snapshot partition without adding a temporal term to the detector's objective.

Seed-, query-, and anchor-based methods retrieve nearby communities [57]–[59]. OLCPM maintains overlapping communities online [60]. ComNetX maintains a complete, nonoverlapping partition and keeps community assignments outside the update region unchanged.

D. ATTRIBUTED AND REPRESENTATION-LEARNING METHODS

Attributed community detection combines topology with vertex features [61]. Among methods based on graph neural networks [62], SEAL learns detection heuristics [63], CommDGI uses Deep Graph Infomax [64], and DMoN and DGCluster use neural modularity objectives [7], [65]. MAGI connects modularity maximization with contrastive learning [8]. Comparative studies and surveys examine these choices of representations, losses, and evaluation criteria [66], [67].

DAG learns communities without a prescribed cluster count [68]. Other methods use deep structural entropy [69] or incorporate group-sensitive connectivity in fairness-aware clustering [70]. Work on efficiency includes pre-training followed by refinement [71] and inductive graph partitioning [72]. We use Spectral Subspace Clustering for Attributed Graphs (S^2CAG) [9] because it can take the quotient adjacency and the features aggregated over its vertices as inputs.

III. PROBLEM FORMULATION†

A. BATCHED GRAPH MODEL

Let $G_t = (V, E_t, \boldsymbol{A}_t, \boldsymbol{X})$ be the graph after batch t, with a fixed vertex set V, $n = |V|$, sparse adjacency $\boldsymbol{A}_t$, and optional node-feature matrix $\boldsymbol{X} \in \mathbb{R}^{n \times d}$. The model permits edge insertions and deletions [73]. A sparse signed batch $\boldsymbol{\Delta}_t$ updates the adjacency as

$$\boldsymbol{A}_t = \boldsymbol{A}_{t-1} + \boldsymbol{\Delta}_t. \tag{1}$$

The affected endpoints are

$$S_t = \{i : \exists j,\ \Delta_t(i,j) \neq 0 \text{ or } \Delta_t(j,i) \neq 0\}. \tag{2}$$

We require $\boldsymbol{A}_t$ to remain nonnegative. The experiments use insertions and positive weight increments with fixed vertex attributes.

A compatible detector returns a partition through the interface

$$\mathcal{B}(\boldsymbol{A}, \boldsymbol{X}, \boldsymbol{y}_0; \boldsymbol{\theta}) \longmapsto \boldsymbol{y}, \tag{3}$$

where $\boldsymbol{X}$ and the initial partition $\boldsymbol{y}_0$ are optional, and $\boldsymbol{\theta}$ contains settings such as resolution or an iteration budget. The output has one integer label per input vertex. Vertices with the same label form a community. A native dynamic

implementation can serve as a backend if it can partition each supplied quotient independently.

Full-snapshot recomputation supplies the entire adjacency $\boldsymbol{A}_t$ and, when supported, the preceding partition:

$$\boldsymbol{y}_t^{\text{full}} = \mathcal{B}(\boldsymbol{A}_t, \boldsymbol{X}, \boldsymbol{y}_{t-1}^{\text{full}}; \boldsymbol{\theta}). \tag{4}$$

B. PARTITION QUALITY

The paired comparisons use the same undirected representation of each graph, symmetrizing directed source graphs. For a weighted adjacency, let $W = \sum_{i,j} A_{ij} > 0$, $d_i^{\text{out}} = \sum_j A_{ij}$, and $d_j^{\text{in}} = \sum_i A_{ij}$. For a partition C, we compute directed modularity at resolution γ as

$$Q_\gamma(\boldsymbol{A}, C) = \frac{1}{W} \sum_{i,j} \left(A_{ij} - \gamma \frac{d_i^{\text{out}} d_j^{\text{in}}}{W} \right) \mathbf{1}\{C(i) = C(j)\}, \tag{5}$$

which reduces to the standard undirected expression when $\boldsymbol{A}$ is symmetric and $d_i^{\text{out}} = d_i^{\text{in}} = d_i$. The main experiments use $\gamma = 1$. We also use normalized mutual information (NMI),

$$\text{NMI}(C, Z) = \frac{2I(C; Z)}{H(C) + H(Z)}, \tag{6}$$

where I denotes mutual information, H denotes entropy, and Z contains the reference class labels [35].

IV. COMNETX METHOD†

A. UPDATE PROCEDURE AND STORED STATE

The *atoms* at level ℓ are the indivisible blocks of a partition $\mathcal{G}_t^\ell$ of the update region U_t^ℓ. The detector receives their quotient adjacency $\bar{\boldsymbol{A}}_\ell$. The update regions are fixed before the first pass and processed from fine to coarse (Fig. 1).

ComNetX stores the sparse adjacency $\boldsymbol{A}_t$, optional features $\boldsymbol{X}$, and an $L \times n$ array $\boldsymbol{C}_t = (C_t^0, \ldots, C_t^{L-1})$ of integer community labels. Entry $C_t^\ell(v)$ is the label of vertex v at level ℓ after batch t. Let $\mathcal{P}_t^\ell$ be the corresponding partition. We write $\mathcal{P} \preceq \mathcal{Q}$ if every block of $\mathcal{P}$ lies in a block of $\mathcal{Q}$. The finest partition, $\mathcal{P}_t^0$, is reported. The coarser partitions determine contraction groups for later batches. Each block H has the canonical label $\kappa(H) = \min H$.

In the maintenance experiments, Leiden initializes $\mathcal{P}_0^0$ on the full initial graph, including for feature-aware detectors. Each coarser partition is obtained by partitioning the quotient of the preceding level and lifting its labels to V. This gives a nested initial hierarchy in which adjacent levels may coincide. The growth control initializes all levels with singleton blocks.

B. SELECTING THE UPDATE REGIONS

After applying a batch, ComNetX expands S_t by r hops on the support of $\boldsymbol{A}_t + \boldsymbol{A}_t^\mathsf{T}$. Nonnegativity ensures that the sparse matrix–vector traversal computes reachability on this symmetrized support. Let

$$B_t = N_r(S_t) = \{v : \text{dist}(v, S_t) \le r\}. \tag{7}$$

Before any partition is modified, an update region is computed at each level from the previous hierarchy:

$$U_t^\ell = \bigcup_{\substack{H \in \mathcal{P}_{t-1}^\ell \\ H \cap B_t \neq \varnothing}} H, \qquad 0 \le \ell < L. \tag{8}$$

Each finer block meeting B_t belongs to a coarser block also meeting B_t. Hence

$$B_t \subseteq U_t^0 \subseteq U_t^1 \subseteq \cdots \subseteq U_t^{L-1}. \tag{9}$$

C. CONTRACTION AND PARTITION UPDATE

1) Atoms and Quotient Construction

The first pass, at the reported level 0, uses the base atoms

$$\begin{aligned} \mathcal{G}_t^0 = {} & \{\{v\} : v \in B_t\} \\ & \cup \big\{(H \cap U_t^0) \setminus B_t : H \in \mathcal{P}_{t-1}^{L-1}, \\ & \qquad (H \cap U_t^0) \setminus B_t \neq \varnothing\big\}. \end{aligned} \tag{10}$$

Thus vertices in B_t can be reassigned individually. Each remaining atom may span several fine communities within one pre-update coarsest community. At every coarser pass, the atoms are the complete blocks of the already updated finer partition within the update region:

$$\mathcal{G}_t^\ell = \{H \in \mathcal{P}_t^{\ell-1} : H \subseteq U_t^\ell\}. \tag{11}$$

Every pass starts from a fresh restriction of the stored adjacency,

$$\boldsymbol{A}_\ell^{\text{work}} = \boldsymbol{D}_\ell \boldsymbol{A}_t \boldsymbol{D}_\ell, \tag{12}$$

where $\boldsymbol{D}_\ell = \text{diag}(\mathbf{1}_{U_t^\ell})$ zeros rows and columns outside U_t^ℓ. Restricting the indices to U_t^ℓ gives the induced adjacency $\boldsymbol{A}_t[U_t^\ell]$. Let $k_\ell = |\mathcal{G}_t^\ell|$. In the membership matrix $\boldsymbol{P}_\ell \in \{0,1\}^{k_\ell \times n}$, the column for each vertex in U_t^ℓ has a single unit entry in the row of its atom. Columns outside U_t^ℓ are zero. The quotient adjacency is

$$\bar{\boldsymbol{A}}_\ell = \boldsymbol{P}_\ell \boldsymbol{A}_\ell^{\text{work}} \boldsymbol{P}_\ell^\mathsf{T}. \tag{13}$$

The backend therefore receives one vertex per atom and the total weight between every ordered pair of atoms.

For a feature-aware backend, ComNetX supports sum and mean aggregation. For $G_{\ell a} \in \mathcal{G}_t^\ell$,

$$\bar{\boldsymbol{x}}_{\ell a} = \begin{cases} \displaystyle\sum_{i \in G_{\ell a}} \boldsymbol{x}_i, & \text{sum}, \\ \displaystyle\frac{1}{|G_{\ell a}|} \sum_{i \in G_{\ell a}} \boldsymbol{x}_i, & \text{mean}. \end{cases} \tag{14}$$

2) Lifting the Partition

The detector partitions the quotient without initial labels. Each atom's vertices inherit its returned label through $\boldsymbol{P}_\ell$. Each resulting block $H \subseteq U_t^\ell$ receives the canonical label $\min H$. Assignments outside U_t^ℓ remain unchanged (Algorithm 1).

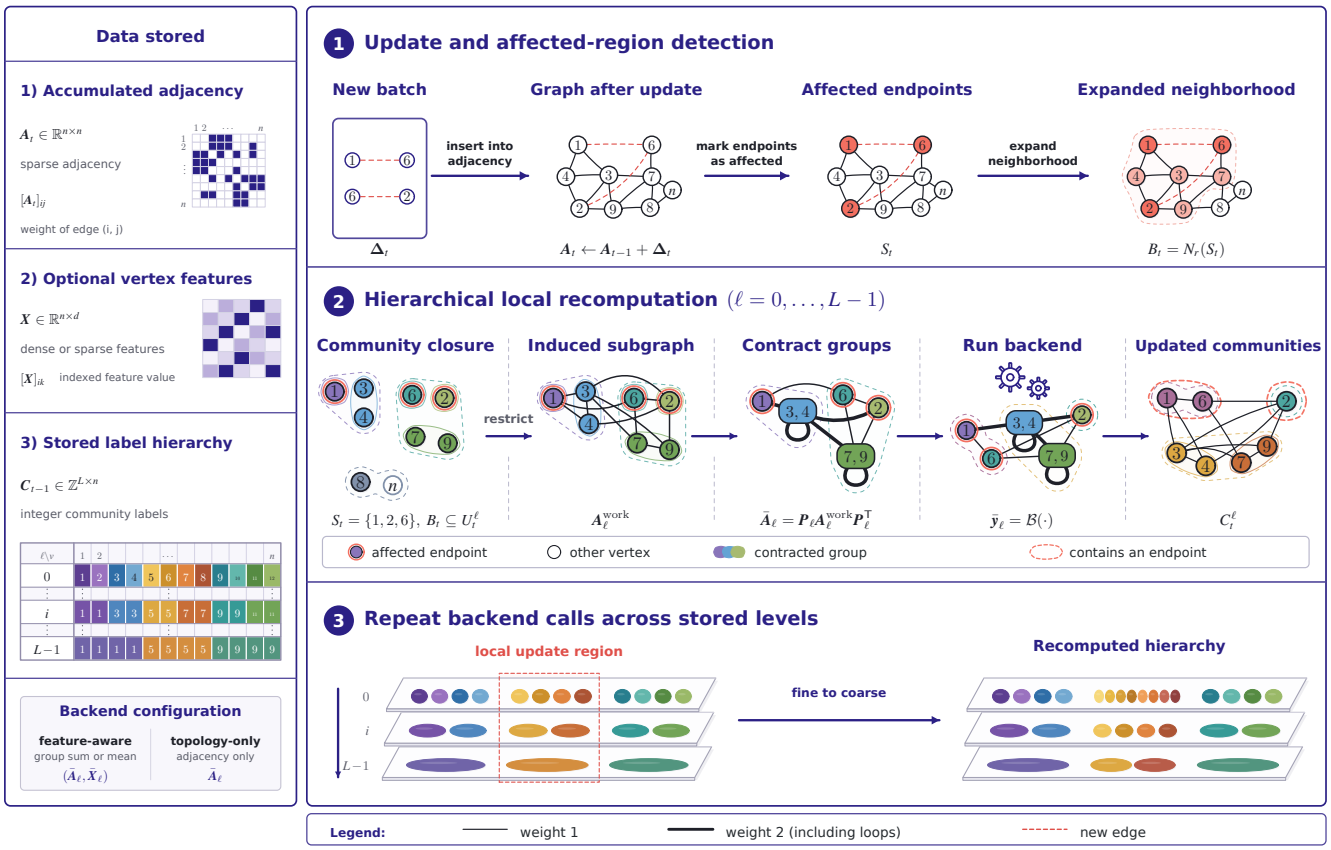


FIGURE 1. **Local recomputation after insertion of edges (1, 6) and (6, 2) (Algorithm 1). The middle panel shows the operations at each level $\ell = 0, \ldots, L-1$. The nested outlines in the closure diagram show the pre-update hierarchy. At level 0, (10) gives each vertex in B_t its own atom and groups the remaining vertices of U_t^0 by their coarsest pre-update community. At each coarser level, (11) contracts each complete block of the updated finer partition within U_t^ℓ to one quotient vertex. In the coarser pass illustrated here, the detector merges the quotient vertices for $\{3, 4\}$ and $\{7, 9\}$, giving all four original vertices the same label. Vertices 8 and n lie outside the update region and retain their labels. The stored arrays and the hierarchy in the lower panel are schematic.**

D. STRUCTURAL PROPERTIES

1) Hierarchy Consistency

Starting from a canonical, nested hierarchy, Algorithm 1 preserves nesting:

$$\mathcal{P}_t^0 \preceq \mathcal{P}_t^1 \preceq \cdots \preceq \mathcal{P}_t^{L-1}. \tag{15}$$

For $\ell > 0$, the pre-update region U_t^ℓ is a union of complete blocks at level $\ell - 1$ by nesting. The finer pass changes only $U_t^{\ell-1} \subseteq U_t^\ell$ by (9). Its newly formed blocks stay inside $U_t^{\ell-1}$, and their canonical labels are disjoint from those of the unchanged outside blocks. Thus U_t^ℓ remains a union of complete updated finer blocks. By (11), each is indivisible at level ℓ. The outside blocks retain their previous nesting. Induction over ℓ proves (15).

2) Modularity under Contraction

We apply the standard modularity-preserving contraction [74] to a region U. Let A_U be its induced adjacency with total weight $W_U > 0$. For a partition $\mathcal{G} = \{G_1, \ldots, G_k\}$ of U with membership matrix $P_U \in \{0,1\}^{k\times|U|}$, the quotient $\bar{A} = P_U A_U P_U^\top$ retains all self-loops and has entries

$$\bar{A}_{ab} = \sum_{i\in G_a} \sum_{j\in G_b} A_{ij}. \tag{16}$$

Consequently, the quotient preserves total weight and the sums of outgoing and incoming weighted degrees. Every quotient partition Z induces a unique partition C of U that keeps each atom intact, and, under the same directed or symmetric adjacency, self-loop, and resolution conventions,

$$Q_\gamma(\bar{A}, Z) = Q_\gamma(A_U, C). \tag{17}$$

The identity follows by summing the internal-weight and degree terms in (5) over the atoms of each community.

The induced graph omits crossing edges and uses a local normalization, so its modularity ranking can differ from the full-graph ranking. Appendix A gives the exact difference. We evaluate the reported finest partition on the full graph.

E. TIME AND MEMORY COMPLEXITY

For $r = 1$, let $d_t(v)$ denote the degree of v in the symmetrized support graph. Then

$$|B_t| \le |S_t| + \sum_{v\in S_t} d_t(v). \tag{18}$$

Algorithm 1 ComNetX update for one batch

Require: A_{t-1}, C_{t-1}, optional X, update Δ_t, radius r, backend $\mathcal{B}$, settings θ

$A_t \leftarrow A_{t-1} + \Delta_t$
$S_t \leftarrow$ endpoints of the updated edges; $B_t \leftarrow N_r(S_t)$
for $\ell = 0, \ldots, L-1$ **do**
  $U_t^\ell \leftarrow$ closure of B_t under $\mathcal{P}_{t-1}^\ell$ ▷ fixed before all passes
end for
$C_t \leftarrow C_{t-1}$
for $\ell = 0, \ldots, L-1$ **do** ▷ fine to coarse
  if $\ell = 0$ **then**
    $\mathcal{G}_t^\ell \leftarrow$ singleton atoms for B_t and the nonempty residual groups in (10)
  else
    $\mathcal{G}_t^\ell \leftarrow$ complete updated $\mathcal{P}_t^{\ell-1}$-blocks in U_t^ℓ
  end if
  $A_\ell^{\text{work}} \leftarrow D_\ell A_t D_\ell$
  build P_ℓ from $\mathcal{G}_t^\ell$
  $\bar{A}_\ell \leftarrow P_\ell A_\ell^{\text{work}} P_\ell^{\mathsf{T}}$
  aggregate $\bar{X}_\ell$ if $\mathcal{B}$ uses features
  $\bar{y}_\ell \leftarrow \mathcal{B}(\bar{A}_\ell, \bar{X}_\ell, \varnothing; \theta)$
  lift $\bar{y}_\ell$ to U_t^ℓ; set $C_t^\ell(v) \leftarrow \min H$ for every lifted block H and $v \in H$
end for
return C_t with reported partition C_t^0

Thus, a small edge batch can still have a large radius-one expansion when it touches hubs. If $\mathcal{P}_{t-1}^\ell$ is the pre-update partition at level ℓ, the update-region size $h_\ell = |U_t^\ell|$ is exactly

$$h_\ell = \sum_{H \in \mathcal{P}_{t-1}^\ell : H \cap B_t \neq \varnothing} |H|. \tag{19}$$

Let $r_t \le r$ be the number of neighborhood traversals, s_t the number of stored adjacency entries, e_ℓ the number of entries in $A_t[U_t^\ell]$, and k_ℓ and q_ℓ the numbers of quotient vertices and stored entries in the quotient adjacency matrix. Let $\overline{T}_{\mathcal{B}}(k,q,d)$ and $\overline{M}_{\mathcal{B}}(k,q,d)$ be nondecreasing worst-case envelopes for the core backend time and working memory. We account for format conversion separately, assuming its cost is linear in the stored adjacency and feature arrays.

1) Worst-Case Resources

Let $F = nd$ bound dense or sparse feature storage for feature-aware calls and $F = 0$ for topology-only calls. Suppose levels are processed sequentially, with quotient edges constructed by direct endpoint relabeling followed by comparison-based sort-and-reduce. After applying the batch, an update requires at most

$$T_{\text{update}} = O\Big(r_t(n+s_t) + L\big[(n+s_t)\log(2+n+s_t) + F\big] + \sum_{\ell=0}^{L-1} \overline{T}_{\mathcal{B}}(k_\ell, q_\ell, d)\Big) \tag{20}$$

time and

$$M_{\text{peak}} = O\Big(s_t + Ln + F + \max_\ell\{e_\ell + h_\ell + q_\ell + \overline{M}_{\mathcal{B}}(k_\ell, q_\ell, d)\}\Big) \tag{21}$$

memory. The time bound includes r_t sparse traversals and, at each level, scans of the label and edge arrays, sorting at most s_t entries, feature aggregation, format conversion, and a backend call. Beyond the $O(s_t + Ln + F)$ stored state, only one level and its backend workspace are needed at a time.

Since $h_\ell, k_\ell \le n$ and $e_\ell, q_\ell \le s_t$, a nondecreasing backend cost gives the explicit graph-wide worst case

$$T_{\text{worst}} = O\Big(r_t(n+s_t) + L\big[(n+s_t)\log(2+n+s_t) + F + \overline{T}_{\mathcal{B}}(n, s_t, d)\big]\Big). \tag{22}$$

An update touching a hub or a large community can make each of the L quotient graphs comparable in size to the full snapshot.

2) Costs of the Present Implementation

The PyTorch implementation uses sparse tensor operations in place of the sort-and-reduce construction assumed in (20). It scans the full label and edge arrays at each level. When it executes r_t traversals and all L passes, it incurs

$$\Omega(r_t s_t + L(n+s_t)) \tag{23}$$

full-scan memory traffic even for small quotients. Appendix B gives the temporary-memory requirements.

V. EXPERIMENTAL DESIGN†

RQ1 What quality–time trade-off does ComNetX achieve relative to warm-started full-snapshot recomputation?
RQ2 How does this trade-off change with a feature-aware detector or a specialized dynamic detector?
RQ3 How large are the expanded neighborhood, the update region, and the quotient graph supplied to the detector?
RQ4 How sensitive is the measured quality–time trade-off to hierarchy depth, radius, feature representation, aggregation, and random-feature initialization?
RQ5 How does the trade-off evolve over longer update sequences, and how does it depend on update granularity and the fraction of the graph already available at initialization?

TABLE 1. Source graphs and actual updates under 999:10. Source counts are undirected edges for Cora, ACM, and CiteSeer, and directed arcs for the other graphs.

Dataset	Vertices	Source count	Order	Updates
Cora	2,708	5,278	artificial	10
ACM	3,025	13,128	artificial	10
CiteSeer	3,327	4,552	artificial	8
Patent	12,214	41,916	timestamp	10
PubMed	19,717	44,338	artificial	10
ArxivMath	270,013	799,745	timestamp	10

A. DATASETS AND STREAM CONSTRUCTION

We use six graphs (Table 1). Patent and ArxivMath supply timestamped records, which we process in chronological order, and 128-dimensional sinusoidal encodings of vertex IDs as node features [75], [76]. Cora, ACM, CiteSeer, and PubMed are static attributed graphs, so we use a fixed artificial order for each stream. The first three are available from the collection described in [67], and the Cora and CiteSeer topologies match the preprocessed Planetoid graphs [77]. For PubMed we use the 44,338-arc version distributed by PyTorch Geometric.[1]

We form batches from stored records. Cora, ACM, and CiteSeer store both orientations of each edge. When they arrive in different batches, the first record inserts the undirected edge and the second increases its weight. Full and local runs use the same weighted snapshots.

In protocol $p{:}q$, we divide the stream into $p+1$ roughly equal parts, use the first p to build the initial graph, and process the last part in up to q batches. Thus $999{:}10$ tests maintenance after approximately 99.9% of the records have arrived. Varying q changes the batch size over the same tail. The $99{:}50$ and $9{:}500$ protocols reserve larger tails (about 1% and 10%, respectively), while $0{:}10$ measures growth from singleton partitions over the entire stream.

The Leiden comparisons in Table 2 cover graphs with paired results. The short maintenance series lacks a Patent run, and the growth series lacks a ComNetX run on ArxivMath.

B. TIMING AND COMPUTING ENVIRONMENT

We report cumulative processing time over the update stream (Table 3). For full recomputation and ComNetX, we measure elapsed time and subtract the recorded format conversion and preprocessing time. For native DF-Leiden, we first submit the batch, then use the time reported by the implementation to update the partition. The time ratio is baseline time divided by ComNetX time; for DF-Leiden the two measurements therefore have different boundaries.

The original experiments ran on a Linux server with two 64-core AMD EPYC 7742 central processing units (CPUs), 2.0 TiB of memory, and eight 80 GB NVIDIA A100 graphics processing units (GPUs). The repeated S^{2}CAG experiments used the same processor model and one visible A100. Graph tensors resided on the GPU and the spectral solver used eight processor threads.

We measured graph sizes separately on a CPU for Cora, PubMed, and ArxivMath using ComNetX with Leiden $(999{:}10, L=3, r=1)$. All timing comparisons use the server runs.

C. METHODS AND FIXED CONFIGURATION

For Leiden [6], we use the Python package *leidenalg*[2] to optimize weighted modularity at resolution one, with two iterations and random seed 1. Full recomputation starts from its own preceding partition; Leiden on each ComNetX quotient starts from singleton communities. By default, ComNetX uses $L=3$ and $r=1$.

DF-Leiden [14] expands the set of vertices to reconsider as community assignments change. We used a Python binding to the authors' OpenMP implementation.[3] The native baseline retains its state between batches. ComNetX starts a fresh instance on each quotient.

1) S^{2}CAG Solver Settings

We use the spectral subspace clustering method of Lin et al. [9], with code adapted from the authors' implementation.[4] The propagation order is $T=10$ with weight $\alpha=1$. The randomized singular value decomposition (SVD) uses $\tau=7$ power iterations, ten oversampling columns, and seed 0; rounding takes 100 iterations. Only the modularity-based variant (M-S^{2}CAG) takes a resolution parameter. We use S^{2}CAG and evaluate modularity at resolution one. Each invocation runs the solver five times and returns the last partition. The internal quality scores use that partition alone and do not measure variation across the five calls.

2) Requested and Realized Cluster Counts

For full S^{2}CAG, the number of communities in the preceding partition sets the requested cluster count k. The previous labels also serve as a reference for internal scoring; neither the spectral solver nor the rounding step uses them as an initial assignment. A local call requests one cluster per quotient vertex. For n vertices and d features, the embedding discards the leading left singular vector and retains up to k of the remaining vectors. Its dimension is thus $s \le \min\{k, \min(n,d)-1\}$. Rounding can leave groups empty, so a call may return fewer than s communities. The reported ComNetX partition also includes the communities outside the update region.

3) Feature Inputs and Preprocessing

ComNetX uses mean aggregation unless specified otherwise. For the random features, we draw 64 independent standard normal entries per vertex once per graph on the CPU. The default seed is 42. The feature-seed study uses seeds 1–5.

[1] https://pytorch-geometric.readthedocs.io/en/latest/generated/torch_geometric.datasets.AttributedGraphDataset.html

[2] https://github.com/vtraag/leidenalg

[3] https://github.com/puzzlef/leiden-communities-openmp-dynamic

[4] https://github.com/HKBU-LAGAS/S2CAG

TABLE 2. Experiments under 999:*q*, 99:50, 9:500, and 0:10. Each original timing comparison uses one execution per method and setting within its series. The repeated S²CAG study uses five fixed-input pairs per setting.

Purpose	Graphs	Protocol	Actual updates	Runs and controls
Main Leiden comparison	five real topologies	999:10	8–10	one paired execution
Feature-aware backend	six real topologies	999:10	8–10	dataset/random features
Repeated S²CAG pairs	PubMed, ArxivMath	999:5, 999:10	5, 10	five fixed-input repeats
Topology parameters	Cora, PubMed, ArxivMath	999:10	10	12 settings per graph
Graph-size diagnostics	Cora, PubMed, ArxivMath	999:10	10	instrumented runs
S²CAG feature controls	Cora, PubMed	999:10	10	modes, aggregation, radius
Feature-input sensitivity	six real topologies	999:10	8–10	feature seeds 1–5
Batch granularity	PubMed, ArxivMath	999:*q*	5–320	seven requested chunk counts
Larger update windows	PubMed, ArxivMath	99:50, 9:500	50, 500	one edge order per graph
Graph growth	five paired real topologies	0:10	10	singleton initialization

TABLE 3. Timing boundaries for all *p*:*q* protocols. "Excluded" means outside the timer or explicitly subtracted. In the graph-growth experiments there is no separate initialization step; processing the first snapshot is included.

Stage	Full	ComNetX	Native DF
Data loading and initial feature generation	Excluded	Excluded	Excluded
Initial partition construction	Excluded	Excluded	Excluded
Applying the batch (full and ComNetX) or submitting it (native DF)	Excluded	Excluded	Excluded
Neighborhood expansion and community closure	–	Included	–
Induced subgraph, quotient, aggregation, lifting	–	Included	–
Adapter conversion/preprocessing interval	Subtracted	Subtracted	–
Detector computation	Included	All passes	Reported time[a]
S²CAG densification and five solver calls	Included	Included	–
S²CAG internal accuracy, NMI, and adjusted Rand index	Included	Included	–
External modularity and reference-label metrics	Excluded	Excluded	Excluded

[a]Native DF-Leiden reports one interval after batch submission. It does not report graph maintenance and partition optimization separately.

Local S²CAG calls first aggregate signed features over atoms, then apply elementwise absolute value, followed by term frequency–inverse document frequency (TF–IDF) weighting and row L2 normalization. Positive and negative values can therefore cancel inside an atom before the absolute value is taken. Full calls apply the absolute value directly to the original features. The one-hot control supplies an identity feature matrix, whose rows are aggregated using the same atom membership.

4) Repeated Paired Experiments

A separate campaign evaluates PubMed and ArxivMath under 999:5 and 999:10, using five full/ComNetX pairs per setting. It uses the same fixed S²CAG solver settings, 64-dimensional random features with seed 42, and ComNetX with $L = 3$, $r = 1$, and mean aggregation. A shared initial hierarchy is constructed with *leidenalg* 0.12.0 using the Leiden settings above. The same initial hierarchy and record sequence are reused across the two subdivisions and all repeats. The initial finest partitions contain 55 communities on PubMed and 8,167 on ArxivMath.

Within each pair the methods run in separate processes, with their order alternating across pairs. Timing starts after warm-up and excludes batch application, external evaluation, and writing labels. We synchronize the device at the interval boundaries and report time both with and without adapter preprocessing. Both intervals include densification, the five solver calls, and their internal metrics.

VI. RESULTS AND DISCUSSION[†]

A. QUALITY AND PROCESSING TIME ON SHORT MAINTENANCE STREAMS (RQ1)

Under 999:10, warm-started full Leiden took 6.60 to 98.72 times as long as ComNetX (Table 4). ArxivMath had the largest time ratio: processing time fell from 110.4 to 1.118 s, and final modularity from 0.8858 to 0.8812.

Across the five graphs, absolute final modularity differences ranged from 0.0035 on CiteSeer to 0.0213 on ACM. Agreement with reference labels decreased on all five.

B. RESULTS WITH DIFFERENT COMMUNITY-DETECTION BACKENDS (RQ2)

Over 500 updates, ComNetX with DF-Leiden was faster than native DF-Leiden on ArxivMath and slower on PubMed (Table 12). With dataset features, ComNetX reduced S²CAG processing time on five of six graphs (Table 5). On ArxivMath, time fell from 4151.6 to 80.270 s, a 51.72-fold reduction. On Patent, time rose from 11.390 to 13.474 s, with gains in modularity and NMI. Across the six graphs, final NMI increased on five.

TABLE 4. Final quality and cumulative measured processing time for full-snapshot and ComNetX Leiden on 999:10 streams.

Dataset	Full Q	Local Q	Full NMI	Local NMI	Full T (s)	Local T (s)	Speedup
Cora	0.8240	0.8141	0.4667	0.4507	0.982	0.149	6.60
ACM	0.8255	0.8042	0.2700	0.2659	0.930	0.123	7.54
CiteSeer	0.8932	0.8897	0.3305	0.3304	0.647	0.077	8.43
PubMed	0.7872	0.7822	0.2067	0.1983	4.727	0.157	30.12
ArxivMath	0.8858	0.8812	0.4226	0.4070	110.4	1.118	98.72

TABLE 5. Main S^{2}CAG feature comparison under 999:10; quality and time values are full-snapshot/ComNetX. Local calls use $L = 3, r = 1$ and mean aggregation. Each setting has one paired execution.

Dataset	Features	Q	NMI	T (s)	Time ratio
Cora	dataset	0.6951/0.7859	0.4355/0.4504	29.949/5.553	5.39
ACM	dataset	0.0536/0.6219	0.0814/0.2538	224.2/10.713	20.93
CiteSeer	dataset	0.7770/0.7719	0.2542/0.3262	135.8/1.792	75.83
Patent	dataset	0.3970/0.5520	0.0604/0.3560	11.390/13.474	0.85
PubMed	dataset	0.6221/0.4987	0.1686/0.1524	78.869/13.609	5.80
ArxivMath	dataset	0.0003/0.0844	0.0002/0.1410	4151.6/80.270	51.72
Cora	random	-0.0020/0.7877	0.0233/0.4505	9.408/4.357	2.16
ACM	random	0.0004/0.6222	0.0089/0.2567	11.286/5.256	2.15
CiteSeer	random	-0.0031/0.7719	0.0170/0.3262	9.519/0.073	129.81
Patent	random	-0.0055/0.6579	0.0009/0.3560	10.165/8.802	1.15
PubMed	random	0.0027/0.5010	0.0003/0.1580	18.272/6.199	2.95
ArxivMath	random	0.0012/0.5875	0.0002/0.1769	3889.9/63.597	61.16

With random features, full S^{2}CAG returned partitions with near-zero modularity on all six graphs. ComNetX obtained higher values using the stored hierarchy. The time ratios include the effect of the different cluster counts requested on the full and quotient graphs.

1) Repeated S^{2}CAG Comparisons

ComNetX was faster in all 20 pairs under both timing definitions (Table 6). With preprocessing subtracted, median speedups are 3.44 and 3.37 on PubMed and 37.63 and 70.40 on ArxivMath for 999:5 and 999:10, respectively. Including preprocessing gives 3.52, 3.46, 37.64, and 70.34 in the same order.

On ArxivMath, the first full-graph update requests $k = 8167$ and returns 16 and 15 groups under 999:5 and 999:10, respectively. It accounts for 93.64% and 90.85% of cumulative full processing time. The implemented randomized SVD uses $k + 10$ projection columns before truncating the retained embedding to at most 63 columns. The aggregate speedups therefore depend strongly on the adapter's initial cluster-count request. As a post hoc timing diagnostic, summing only updates $2, \ldots, q$ for both methods gives median paired speedups of 3.06 and 7.46, respectively (3.12 and 7.65 with preprocessing included).

Full S^{2}CAG returned 11 or 13 communities; ComNetX returned 206–392 on PubMed and over 8,100 on ArxivMath, with higher final modularity and NMI in all four comparisons. On these short streams with random features, keeping the initial Leiden labels fixed gave higher scores than either update: $Q = 0.7812$ and NMI $= 0.2009$ on PubMed, and $Q = 0.8785$ and NMI $= 0.4005$ on ArxivMath, evaluated on the final graph for either subdivision.

Increasing q from 5 to 10 raised mean cumulative ComNetX time from 3.526 to 6.020 s on PubMed and reduced it from 101.379 to 56.371 s on ArxivMath. Most of the ArxivMath reduction occurred in the first pass. Its quotients had 4,327–4,976 vertices at $q = 5$ and 2,128–3,049 at $q = 10$, and mean cumulative solver time for that pass fell from 97.472 to 51.222 s.

C. SIZE OF THE CONTRACTED GRAPHS (RQ3)

On PubMed and ArxivMath, the radius-one neighborhoods averaged 154.9 and 782.0 vertices, respectively (Table 7). Community closure expanded these regions to 64.3% and 78.7% of all vertices on average. After contraction, the first detector call processed just 167.9 and 825.3 quotient vertices on average.

The second and third passes had identical quotient sizes in all 30 updates. The initial hierarchy levels coincided in these runs, and the second pass left the atom count unchanged for the third.

Region selection still scans the full label and edge arrays (Section IV).

D. SENSITIVITY TO HIERARCHY DEPTH, RADIUS, AND FEATURES (RQ4)

ComNetX was faster than full Leiden, with lower final modularity in every case (Fig. 2). Across the 36 depth–radius

TABLE 6. Repeated paired S²CAG experiments under 999:5 and 999:10, with random features, $L = 3$, $r = 1$, and mean aggregation. Q, NMI, and K are full/ComNetX final values; K counts nonempty communities in the reported partition of the entire graph. Each method returns identical Q, NMI, and K in its five repeats at a given setting. Times exclude adapter preprocessing and are mean ± sample standard deviation in seconds. S is the median of the five paired time ratios.

Dataset	Protocol	Q	NMI	K	Full T (s)	ComNetX T (s)	S
PubMed	999:5	0.0001/0.4693	0.0003/0.0869	11/206	11.995±0.101	3.526±0.063	3.44
PubMed	999:10	0.0001/0.4683	0.0003/0.1112	11/392	20.342±0.095	6.020±0.065	3.37
ArxivMath	999:5	0.0020/0.5724	0.0003/0.1909	13/8,119	3819.065±21.885	101.379±0.563	37.63
ArxivMath	999:10	0.0024/0.5633	0.0003/0.1774	13/8,116	3951.633±37.385	56.371±0.196	70.40

TABLE 7. Measured graph dimensions, averaged over ten updates in the separate CPU diagnostics (999:10, $L = 3$, $r = 1$). Parentheses give percentages of all n vertices. Here $h_0 = |U_t^0|$, k_ℓ is the number of contracted vertices at level ℓ, and q_0 counts stored adjacency entries, including self-loops and both orientations of symmetric edges.

Dataset	$\lvert B_t\rvert$	h_0	k_0	q_0	$k_1 = k_2$
Cora	9.4	314.2 (11.6%)	11.6 (0.43%)	45.6	3.0
PubMed	154.9	12,686.8 (64.3%)	167.9 (0.85%)	1,393.9	13.0
ArxivMath	782.0	212,558.8 (78.7%)	825.3 (0.31%)	7,408.8	47.4

configurations, speedups were 8.52–23.42× on Cora, 7.63–125.54× on PubMed, and 15.27–636.84× on ArxivMath.

At radius one, a single pass was faster than three passes on all three graphs. Increasing depth or radius did not consistently improve modularity. On ArxivMath, for example, the fastest setting ($L = 1, r = 0$) gave modularity 0.8792, compared with 0.8819 at $L = 2, r = 2$. We used a common setting ($L = 3, r = 1$) for the comparisons across datasets.

1) Effect of Feature Representation

Tables 8 and 5 use separate runs with the same solver settings. For the feature settings shared by the two series on Cora and PubMed, final Q and NMI agreed to the recorded precision. Each time comparison uses the full and local runs from the same series.

Replacing dataset features with random features changed ComNetX modularity from 0.7859 to 0.7877 on Cora and from 0.4987 to 0.5010 on PubMed, while full S²CAG returned near-zero modularity with random features. With one-hot features on PubMed, ComNetX reduced processing time from 1973.0 to 120.0 s.

2) Feature Aggregation and Radius

The matching quality scores in Table 9 are expected under the S²CAG preprocessing used here. For each nonempty atom a, $\boldsymbol{x}_{\mathrm{mean}}(a) = \boldsymbol{x}_{\mathrm{sum}}(a)/|a|$. Taking absolute values preserves this positive scale factor and leaves the feature supports, hence the IDF weights, unchanged. Linear TF–IDF weighting preserves the factor, and row L2 normalization cancels it; zero rows remain zero. Thus mean and sum yield the same normalized features in exact arithmetic, although floating-point rounding can introduce small differences. Each timing is based on one execution.

In Table 10, radius zero yields the highest modularity on both graphs. Increasing the radius to two reduces PubMed modularity from 0.6601 to 0.1659. Changing the radius alters both the vertices that can be reassigned individually and the features presented to the detector.

3) Variation across Random-Feature Seeds

Across the five random-feature seeds (Table 11), the sample standard deviation of final modularity is below 0.002 on Cora, ACM, CiteSeer, and Patent. It is 0.0379 on PubMed and 0.0665 on ArxivMath. PubMed shows the largest variation in NMI.

E. LONGER STREAMS, UPDATE GRANULARITY, AND INITIALIZATION (RQ5)

1) Sustained Maintenance over 500 Updates

Over 500 updates (9:500), full Leiden took 24.46 times as long as ComNetX on PubMed and 72.67 times as long on ArxivMath (Table 12). The 500 ArxivMath batches span two distinct source timestamps. Compared with full Leiden, ComNetX gave slightly lower final modularity on both graphs. Final NMI rose from 0.2017 to 0.2045 on PubMed and fell from 0.4205 to 0.4006 on ArxivMath.

Fig. 3 shows that the absolute modularity difference remains below 0.008 at every measured update on both graphs. The cumulative time ratio remains above one throughout both trajectories.

ComNetX with DF-Leiden reduces ArxivMath time from 134.4 s to 13.517 s, while increasing final modularity from 0.8551 to 0.8792 and NMI from 0.3884 to 0.3967. On PubMed, a 0.354 s increase in processing time accompanies higher modularity (0.7786 versus 0.7619) and NMI (0.2051 versus 0.1946).

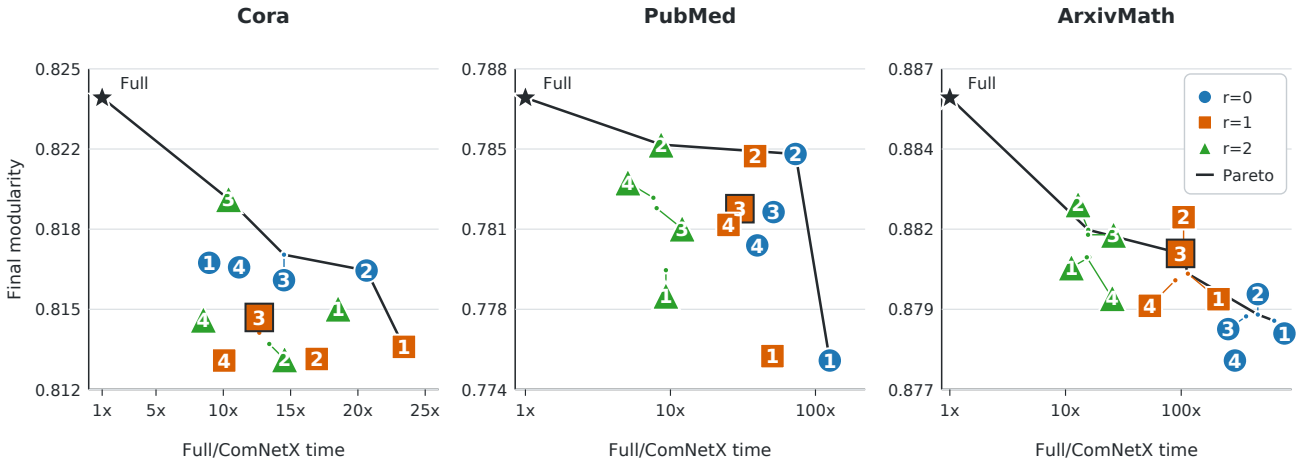


FIGURE 2. Effects of hierarchy depth and radius on Leiden modularity and processing time under 999:10, with $L \in \{1, 2, 3, 4\}$ and radius $r \in \{0, 1, 2\}$. Marker numbers show L; color and shape show r; and the black line connects nondominated measured points. The outlined orange square is the fixed $L = 3, r = 1$ configuration. Each point is one execution; ratios use the full baseline from the same parameter study.

TABLE 8. S²CAG feature sensitivity under 999:10, with ten updates on each graph. Each cell reports Q / NMI / T (seconds).

Variant	Cora	PubMed
Full, dataset features	0.6951/0.4355/30.156	0.6221/0.1686/78.119
ComNetX, dataset features	0.7859/0.4504/5.407	0.4987/0.1524/14.394
Full, random features	-0.0020/0.0233/8.571	0.0027/0.0003/18.861
ComNetX, random features	0.7877/0.4505/4.240	0.5010/0.1580/6.017
Full, one-hot input	0.6974/0.3528/35.310	0.6540/0.1071/1973.0
ComNetX, one-hot input	0.7862/0.4500/5.632	0.5349/0.1884/120.0

TABLE 9. S²CAG feature aggregation under 999:10, at $L = 3, r = 1$, using random features.

Dataset	Aggregation	Q	NMI	T (s)
Cora	mean	0.7877	0.4505	5.929
Cora	sum	0.7877	0.4505	4.197
PubMed	mean	0.5010	0.1580	6.177
PubMed	sum	0.5010	0.1580	5.995

TABLE 10. S²CAG radius sensitivity under 999:10, at $L = 3$, with random features and mean aggregation.

Dataset	r	Q	NMI	T (s)
Cora	0	0.8156	0.4509	1.740
Cora	1	0.7877	0.4505	4.368
Cora	2	0.6541	0.4120	4.829
PubMed	0	0.6601	0.1350	4.386
PubMed	1	0.5010	0.1580	6.248
PubMed	2	0.1659	0.0329	30.304

2) Subdivision of a Fixed Update Window

We divided the same final 0.1% of records into progressively smaller batches (Table 13). At $q = 10$, the separate runs on PubMed gave a Leiden time ratio of 31.42, compared with 30.12 in Table 4, with identical final Q and NMI. On ArxivMath, increasing q from 5 to 320 raised the Leiden time ratio from 64.12 to 695.84 and the final modularity gap from 0.0039 to 0.0134. Full Leiden processed the entire graph after each of the smaller batches.

PubMed has only 44 tail records, so requests for 80, 160, and 320 batches produce the same 44 updates. Final quality is identical, and the measured Leiden time ratios range from 45.43 to 45.57. Across all granularity settings, ComNetX with DF-Leiden has higher final modularity and NMI than the native stream on both graphs; its time ratio is 8.45–10.73 on ArxivMath and below one on PubMed.

3) Increasing the Updated Fraction

The 99:50 control processes approximately 1% of the stored records, between the 0.1% maintenance window and the 10% window of 9:500. Table 14 shows Leiden speedups of 24.16 on PubMed and 60.26 on ArxivMath, with final modularity differences of 0.0074 and 0.0041. DF-Leiden exhibits the same qualitative pattern as at 500 updates: the native stream is slightly faster on PubMed, while ComNetX is faster on ArxivMath and improves both quality measures on both graphs. Since both the initial snapshot and the number of calls change between protocols, these comparisons do not isolate update-window size as a single causal factor.

TABLE 11. ComNetX with S^{2}CAG and random features under 999:10: mean ± sample standard deviation across feature seeds 1–5 on each fixed graph stream, with $L = 3, r = 1$ and mean aggregation.

Dataset	Q	NMI	T (s)
Cora	0.7853±0.0014	0.4499±0.0012	5.971±0.038
ACM	0.6221±0.0018	0.2571±0.0010	5.229±0.046
CiteSeer	0.7719±0.0000	0.3262±0.0000	1.753±0.004
Patent	0.6576±0.0004	0.3563±0.0002	8.676±0.101
PubMed	0.4481±0.0379	0.1241±0.0197	6.146±0.083
ArxivMath	0.5156±0.0665	0.1671±0.0087	62.490±1.990

TABLE 12. Quality and measured processing time after 500 updates under 9:500. Each pair lists the baseline first and ComNetX second. The baseline is full-graph Leiden initialized with its previous partition or native DF-Leiden with state retained between batches. The time ratio is baseline time divided by ComNetX time.

Dataset	Backend	Q	NMI	T (s)	Time ratio
PubMed	Leiden	0.7864/0.7786	0.2017/0.2045	248.6/10.165	24.46
PubMed	DF-Leiden	0.7619/0.7786	0.1946/0.2051	4.442/4.796	0.93
ArxivMath	Leiden	0.8860/0.8799	0.4205/0.4006	5652.7/77.788	72.67
ArxivMath	DF-Leiden	0.8551/0.8792	0.3884/0.3967	134.4/13.517	9.94

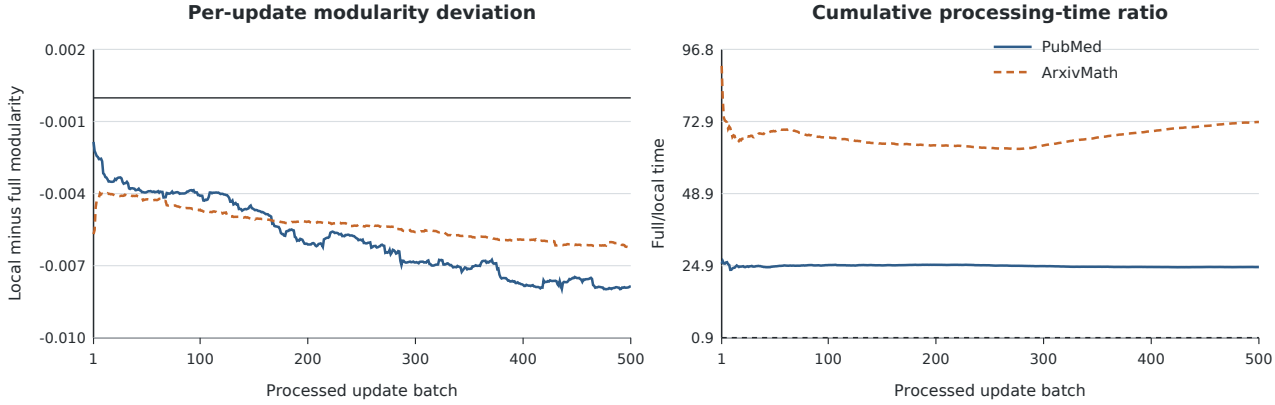


FIGURE 3. Leiden trajectories over 500 updates under 9:500. Left: per-batch modularity difference (ComNetX minus full Leiden), where negative values favor the full partition. Right: cumulative full/ComNetX processing-time ratio, where values above one favor ComNetX.

4) Growth from Singleton Initialization

When both methods started from singleton partitions, ComNetX with Leiden was faster on all five graphs (Table 15). The time ratios of 1.17–1.79 were smaller than for maintenance of an established graph. Final modularity differences ranged from 0.0003 to 0.0080, and NMI increased on Patent and PubMed.

VII. CODE AVAILABILITY AND LIMITATIONS†

The ComNetX implementation, experiment configurations, and archived results are available online.[5] Rerunning S^{2}CAG and native DF-Leiden requires external components not included in the artifact.

[5]https://github.com/mpailab/comnetx/tree/ieee-access-r18-20261109/artifacts/ieee-access

Four of the six graphs use artificial edge orders, and the window controls reuse the same record sequences. Most configurations have one execution per method. The repeated S^{2}CAG study uses five runs per method and setting to assess runtime repeatability at fixed inputs. Independent temporal windows, edge deletions, arriving vertices, and changing node attributes have not been evaluated.

VIII. CONCLUSION†

ComNetX maintains a nested partition by applying an existing detector to weighted quotients of the update regions. With Leiden, the largest time savings occurred when small batches updated an established graph. Over 500 updates on PubMed and ArxivMath, modularity stayed within 0.008 of full Leiden at every step. Against native DF-Leiden, ComNetX was faster on ArxivMath and slower on PubMed, with higher final modularity and NMI on both. For S^{2}CAG, the local updates

TABLE 13. Fixed-window granularity control, 999:*q*. Here *u* is the actual number of nonempty updates, *b* the stored records per update, ΔQ final ComNetX-minus-baseline modularity, and *S* baseline/ComNetX time. DF-Leiden uses its native dynamic baseline. Each row is one execution per method.

Dataset	q	u	b	Leiden ΔQ	Leiden S	DF ΔQ	DF S
PubMed	5	5	8–9	-0.0043	5.24	+0.0206	0.32
PubMed	10	10	4–5	-0.0050	31.42	+0.0207	0.71
PubMed	20	20	2–3	-0.0060	39.21	+0.0277	0.61
PubMed	40	40	1–2	-0.0167	44.83	+0.0076	0.49
PubMed	80	44	1	-0.0179	45.57	+0.0077	0.51
PubMed	160	44	1	-0.0179	45.49	+0.0077	0.51
PubMed	320	44	1	-0.0179	45.43	+0.0077	0.51
ArxivMath	5	5	159–160	-0.0039	64.12	+0.0306	8.45
ArxivMath	10	10	79–80	-0.0046	99.58	+0.0309	10.05
ArxivMath	20	20	39–40	-0.0049	149.57	+0.0354	10.34
ArxivMath	40	40	19–20	-0.0065	232.46	+0.0215	10.71
ArxivMath	80	80	9–10	-0.0080	381.84	+0.0193	10.73
ArxivMath	160	160	4–5	-0.0084	556.49	+0.0207	10.42
ArxivMath	320	320	2–3	-0.0134	695.84	+0.0204	9.99

TABLE 14. Intermediate-window control, 99:50: 50 updates collectively covering approximately 1% of the stored record sequence. Values are baseline/ComNetX, with the same baseline and timing conventions as Table 12.

Dataset	Backend	Q	NMI	T (s)	Time ratio
PubMed	Leiden	0.7883/0.7809	0.2049/0.2052	24.831/1.028	24.16
PubMed	DF-Leiden	0.7401/0.7808	0.1875/0.2068	0.447/0.485	0.92
ArxivMath	Leiden	0.8855/0.8813	0.4181/0.4006	571.3/9.480	60.26
ArxivMath	DF-Leiden	0.8377/0.8810	0.3952/0.4002	13.637/1.506	9.05

TABLE 15. Graph growth under 0:10 from singleton initialization, including the first processed snapshot. Values are full Leiden/ComNetX; each row is one paired execution. The entire stored record sequence is divided into ten batches.

Dataset	Q	NMI	T (s)	Time ratio
Cora	0.8230/0.8150	0.4715/0.4642	0.985/0.640	1.54
ACM	0.8248/0.8232	0.2704/0.2687	0.934/0.801	1.17
CiteSeer	0.8932/0.8860	0.3302/0.3272	0.847/0.473	1.79
Patent	0.8567/0.8564	0.3617/0.3673	2.197/1.351	1.63
PubMed	0.7866/0.7797	0.2014/0.2094	5.652/3.972	1.42

also saved time, particularly when the full detector requested thousands of clusters on its first update.

APPENDIX A BOUNDARY TERM FOR MODULARITY†

To relate modularity on the induced subgraph and on the full graph, assume $A_{ij} \geq 0$, $W = \sum_{i,j} A_{ij} > 0$, and $\gamma \geq 0$. Let $U = U_t^{\ell}$, $W_U = \sum_{i,j\in U} A_{ij} > 0$. Write $\boldsymbol{A}[U] = (A_{ij})_{i,j\in U}$ for the adjacency matrix of the induced subgraph on U, and let Z be a partition of U whose labels are disjoint from the fixed outside partition. Let C_Z combine Z with that retained outside partition. For each block a contained in U, let x_a and y_a be its outgoing and incoming degree within U, and define its boundary degrees by

$$p_a = \sum_{i\in a, j\notin U} A_{ij}, \qquad q_a = \sum_{i\notin U, j\in a} A_{ij}. \tag{24}$$

Let $K_{V\setminus U}$ denote the contribution of the fixed outside communities to full-graph modularity. Substitution in (5) gives

$$\begin{aligned} D(Z) &= Q_\gamma(\boldsymbol{A}, C_Z) - K_{V\setminus U} - \frac{W_U}{W} Q_\gamma(\boldsymbol{A}[U], Z) \\ &= \frac{\gamma}{W^2}\left[\frac{W - W_U}{W_U}\sum_a x_a y_a \right. \\ &\qquad \left. - \sum_a (x_a q_a + y_a p_a + p_a q_a)\right]. \end{aligned} \tag{25}$$

The internal-edge term is unchanged; the mismatch consists of the global-versus-local null-model normalization and the boundary degrees.

Write $\beta_{\text{out}} = \sum_a p_a$ and $\beta_{\text{in}} = \sum_a q_a$. Nonnegativity gives

$$-B_- \leq D(Z) \leq B_+, \tag{26}$$

where

$$\begin{aligned} B_+ &= \frac{\gamma}{W^2} W_U (W - W_U), \\ B_- &= \frac{\gamma}{W^2}\left[W_U(\beta_{\text{out}} + \beta_{\text{in}}) + \beta_{\text{out}}\beta_{\text{in}}\right]. \end{aligned} \tag{27}$$

Both bounds follow from $\sum_a u_a v_a \le (\sum_a u_a)(\sum_a v_a)$ for nonnegative sequences.

For a fixed family $\mathcal{S}$ of partitions that keep the specified atoms intact, define $F(Z) = Q_\gamma(\boldsymbol{A}, C_Z) - K_{V\setminus U}$ and $L(Z) = (W_U/W)Q_\gamma(\boldsymbol{A}[U], Z)$. If Z_U maximizes L and Z_G maximizes F over $\mathcal{S}$, then

$$0 \le F(Z_G) - F(Z_U) \le B_- + B_+. \tag{28}$$

This bound compares exact optima only within that family; it does not bound loss against an unrestricted full-graph optimum or certify the output of a heuristic backend.

APPENDIX B MEMORY USE OF THE CURRENT IMPLEMENTATION[†]

The implementation stores Boolean masks for all fixed update regions and processes levels sequentially. Let $M_{\mathrm{SpMM},\ell}$ and $M_{\mathcal{B},\ell}$ denote the workspace for sparse matrix multiplication and for the detector, respectively, at level ℓ. Let $M_{\mathrm{feat},\ell}$ denote the additional storage for feature preparation and aggregation at that level, with $M_{\mathrm{feat},\ell} = 0$ for topology-only calls. The implementation's total additional space is conservatively bounded by

$$M_{\mathrm{extra}} = O\Big(Ln + \max\big\{s_t + n, \max_\ell \big[e_\ell + h_\ell + q_\ell + M_{\mathrm{feat},\ell} + M_{\mathrm{SpMM},\ell} + M_{\mathcal{B},\ell}\big]\big\}\Big). \tag{29}$$

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GRIGORIY BOKOV received the Candidate of Physical and Mathematical Sciences degree in 2014. He is currently the Director of the AI Research Center, Lomonosov Moscow State University, Moscow, Russia, and an Associate Professor and the Head of the Laboratory of Mathematical Problems of Artificial Intelligence at the Faculty of Mechanics and Mathematics. His research interests include logical calculi, functional systems, automata, machine-learning algorithms, optimization, and large dynamic graphs.

ALEKSANDR KONOVALOV received the Specialist degree from the Faculty of Mechanics and Mathematics, Lomonosov Moscow State University, Moscow, Russia, in 2013, and the Candidate of Physical and Mathematical Sciences degree in 2018. He is currently a Junior Researcher with Lomonosov Moscow State University. His research interests include constructive mathematical logic, graph neural networks, and dynamic community detection.

ANNA UPOROVA is currently pursuing a Specialist degree in the Faculty of Mechanics and Mathematics, Lomonosov Moscow State University, Moscow, Russia. Her research interests include dynamic graphs, community detection, and graph neural networks.

ALEXANDER DROBYSHEV is currently a postgraduate student with the Chair of Mathematical Theory of Intelligent Systems, Faculty of Mechanics and Mathematics, Lomonosov Moscow State University, Moscow, Russia. His research interests include Boolean-network dynamics, limit cycles, computational complexity, and dynamic graph algorithms.

IAROSLAV EGOROV is affiliated with the AI Research Center, Lomonosov Moscow State University, Moscow, Russia. His recent work includes scalable dynamic community detection.